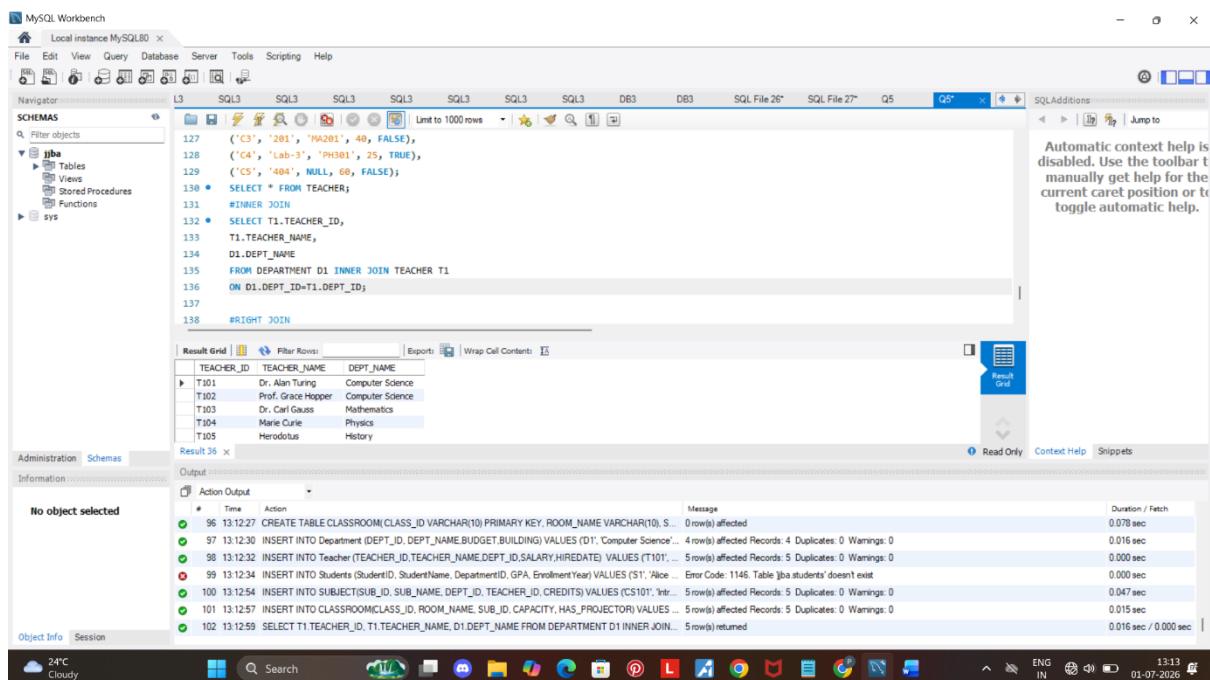


Q. 1 INNER JOIN.

Query:

```
SELECT T1.TEACHER_ID,  
T1.TEACHER_NAME,  
D1.DEPT_NAME  
FROM DEPARTMENT D1 INNER JOIN TEACHER T1  
ON D1.DEPT_ID=T1.DEPT_ID;
```

Screenshot:



Insights:

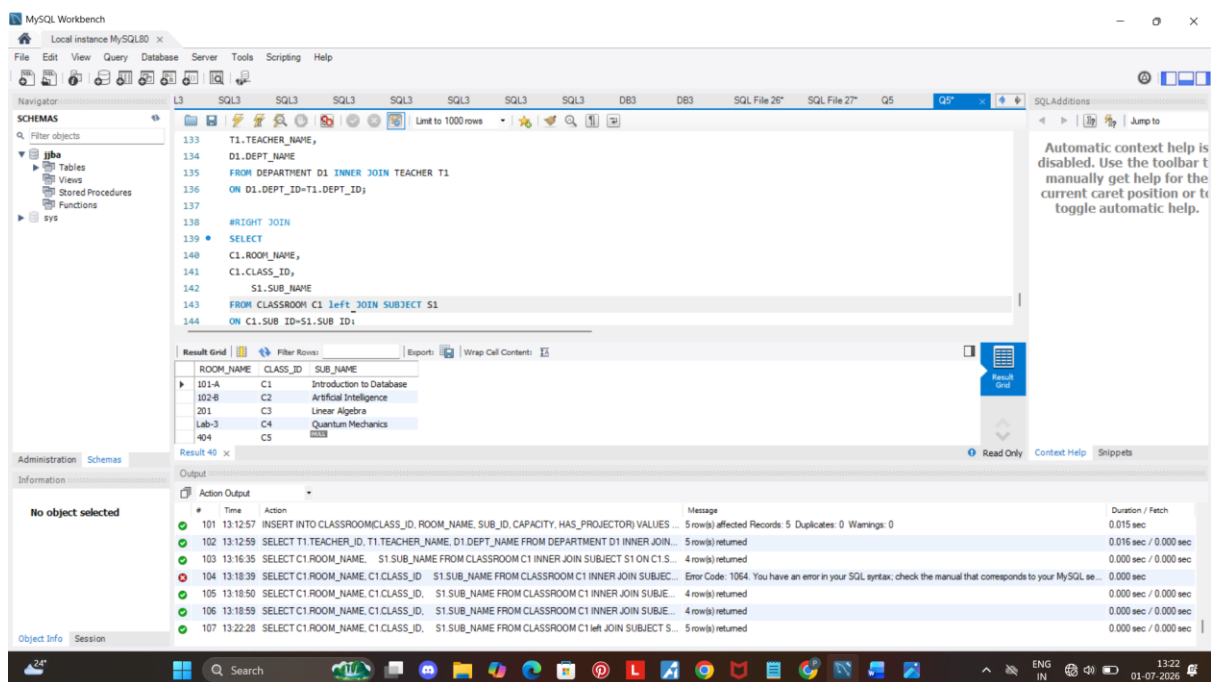
- Fetched the data from teacher and department table with one common column 'DEPT_ID'.
- Based on the DEPT_ID connection we fetched all the common data from both the tables.

Q. 2 LEFT JOIN.

Query:

```
SELECT  
C1.ROOM_NAME,  
C1.CLASS_ID,  
S1.SUB_NAME  
FROM CLASSROOM C1 left JOIN SUBJECT S1  
ON C1.SUB_ID=S1.SUB_ID;
```

Screenshot:



Insight:

- We fetched the common data from table classroom and subject name based on the common column 'DEPT_ID'.
- The row of sub_name is null since it only has 4 subjects in it.

Q. 3 RIGHT JOIN.

Query:

SELECT

C1.ROOM_NAME,

C1.CLASS_ID,

S1.SUB_NAME

FROM CLASSROOM C1 Right join SUBJECT S1

on C1.SUB_ID=S1.SUB_ID;

Screenshot:

The screenshot displays the MySQL Workbench interface. The SQL editor contains the following query:

```
133 T1.TEACHER_NAME,  
134 D1.DEPT_NAME  
135 FROM DEPARTMENT D1 INNER JOIN TEACHER T1  
136 ON D1.DEPT_ID=T1.DEPT_ID;  
137  
138 #RIGHT JOIN  
139 *  
140 SELECT  
141 C1.ROOM_NAME,  
142 C1.CLASS_ID,  
143 S1.SUB_NAME  
144 FROM CLASSROOM C1 Right join SUBJECT S1  
145 on C1.SUB_ID=S1.SUB_ID;
```

The Results grid shows the output of the query:

ROOM_NAME	CLASS_ID	SUB_NAME
101-A	C1	Introduction to Database
102-B	C2	Artificial Intelligence
201	C3	Linear Algebra
Lab-3	C4	Quantum Mechanics

The Output tab shows the execution log with the following messages:

#	Time	Action	Message	Duration / Fetch
105	13:18:50	SELECT C1.ROOM_NAME, C1.CLASS_ID, S1.SUB_NAME FROM CLASSROOM C1 INNER JOIN SUBJE...	4 row(s) returned	0.000 sec / 0.000 sec
106	13:18:59	SELECT C1.ROOM_NAME, C1.CLASS_ID, S1.SUB_NAME FROM CLASSROOM C1 INNER JOIN SUBJE...	4 row(s) returned	0.000 sec / 0.000 sec
107	13:22:28	SELECT C1.ROOM_NAME, C1.CLASS_ID, S1.SUB_NAME FROM CLASSROOM C1 left JOIN SUBJECT S...	5 row(s) returned	0.000 sec / 0.000 sec
108	13:23:43	SELECT C1.ROOM_NAME, C1.CLASS_ID, S1.SUB_NAME FROM CLASSROOM C1 Right JOIN SUBJECT S1 O...	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL se...	0.000 sec
109	13:24:13	SELECT C1.ROOM_NAME, C1.CLASS_ID, S1.SUB_NAME FROM CLASSROOM C1 Right join subject s1 ...	Error Code: 1146. Table 'jba.subject' doesn't exist	0.000 sec
110	13:24:15	SELECT C1.ROOM_NAME, C1.CLASS_ID, S1.SUB_NAME FROM CLASSROOM C1 Right join subject s1 ...	Error Code: 1146. Table 'jba.subject' doesn't exist	0.000 sec
111	13:24:24	SELECT C1.ROOM_NAME, C1.CLASS_ID, S1.SUB_NAME FROM CLASSROOM C1 Right join SUBJECT ...	5 row(s) returned	0.000 sec / 0.000 sec

Insights:

- The rows which doesn't have the subject name is empty.
- It only fetched the data that has common data at the right side.